

House Price Predictor: Linear Regression Report

AI & Machine Learning Internship — Task 1 | California Housing Dataset

Name: [Your Name] **Date:** [Date] **Notebook file:** task1_ml_linear_regression.ipynb

This report is a fill-in template built from the same modeling pipeline as the accompanying notebook. The charts below use example/illustrative data to show you the expected format — **replace them with the actual charts and numbers your notebook produces** when you run it against the real California Housing dataset.

1. Objective

The goal of this task is to build a Linear Regression model that predicts the median house value for districts in California, using features such as median income, house age, average rooms, population, and location. This report summarizes the exploratory data analysis, model training approach, evaluation results, and ideas for improving the model.

2. Dataset

The California Housing dataset (from scikit-learn's *fetch_california_housing*) contains 20,640 records, each representing a California block group. It has 8 numeric input features and one target variable, **MedHouseVal** (median house value, in units of \$100,000).

Feature	Description
MedInc	Median income in the block group
HouseAge	Median house age in the block group
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude / Longitude	Geographic location

3. Exploratory Data Analysis (EDA)

No missing values were found in the dataset (fill in: confirm with *df.isnull().sum()*). The target variable **MedHouseVal** is right-skewed, with most districts valued below \$350,000 and a smaller number of high-value outliers. The correlation heatmap below shows how strongly each feature relates to house value — **MedInc (median income) typically shows the strongest positive correlation.**

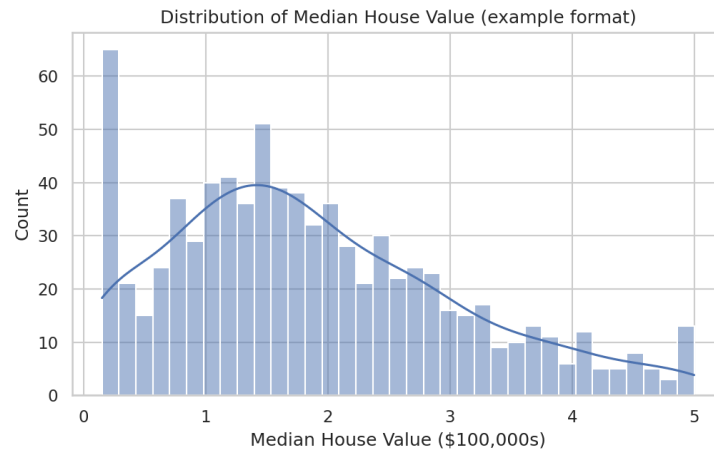


Figure 1 (example format). Replace with your notebook's actual target distribution chart.

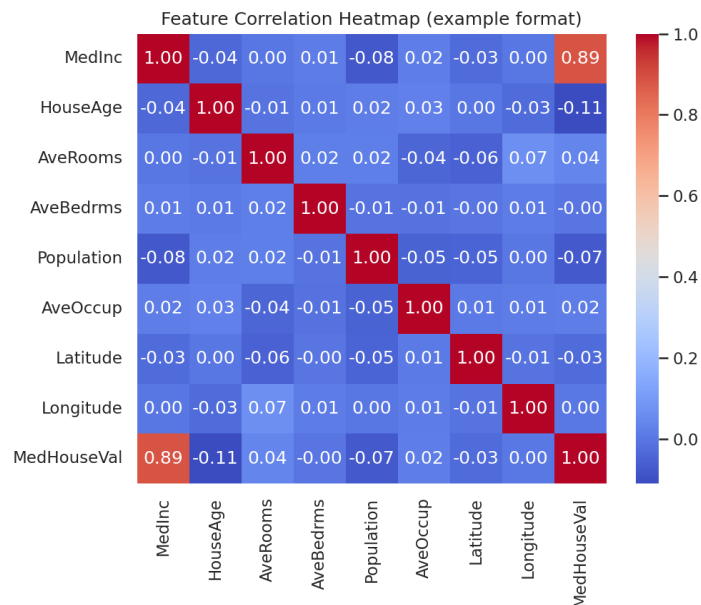


Figure 2 (example format). Replace with your notebook's actual correlation heatmap.

4. Model & Methodology

The data was split into 80% training and 20% testing sets using `train_test_split` (random_state=42). A **Linear Regression** model (scikit-learn) was trained on the 8 input features to predict MedHouseVal. No feature scaling or engineering was applied in this baseline version.

5. Evaluation Metrics

The trained model was evaluated on the held-out test set using three standard regression metrics:

Metric	Your Result	What it means
MAE	[fill in]	Average absolute prediction error, in \$100,000s
RMSE	[fill in]	Similar to MAE, but penalizes large errors more

R^2	[fill in]	Proportion of variance in price explained by the model (0-1)
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For reference, a baseline (unscaled) Linear Regression on this dataset typically achieves an R^2 around 0.6, MAE around 0.5, and RMSE around 0.7 — use this only as a rough sanity check, not a substitute for your own computed values.

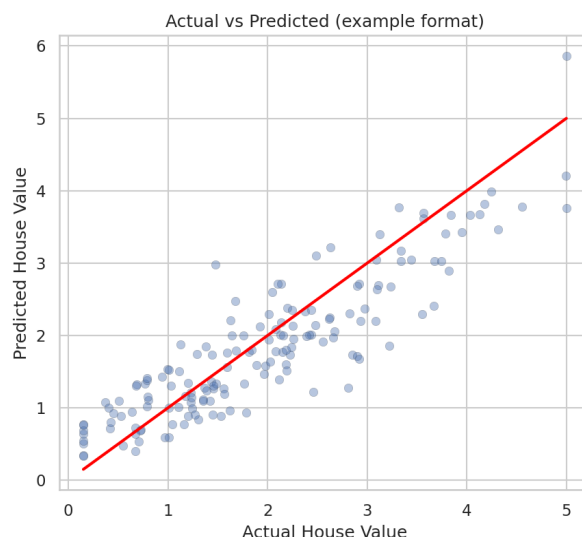


Figure 3 (example format). Replace with your notebook's actual Predicted vs Actual scatter plot.

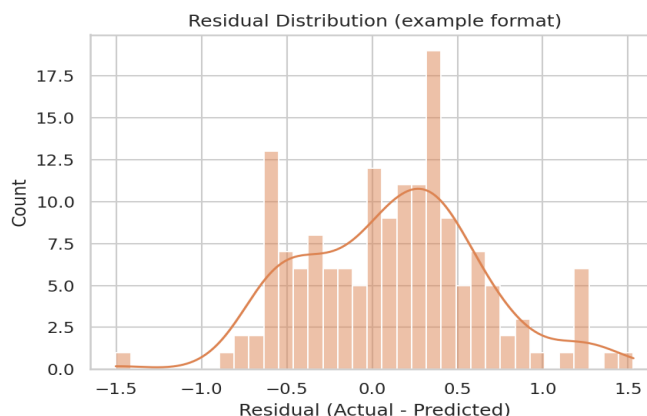


Figure 4 (example format). Replace with your notebook's actual residuals chart.

6. Improvement Ideas

- **Feature engineering** — create ratios like rooms-per-household or bedrooms-per-room; bucket latitude/longitude into regions or distance-to-coast.
- **Handle outliers** — cap or remove extreme values in AveRooms, AveOccup, and Population.
- **Feature scaling** — apply StandardScaler, since features are on very different scales (income vs. population).
- **Try other models** — Ridge/Lasso regression (regularization), Random Forest, or Gradient Boosting typically outperform plain linear regression on this dataset.
- **Cross-validation** — use `cross_val_score` for a more robust performance estimate instead of a single train/test split.

7. Conclusion

[Fill in after running the notebook] The Linear Regression model achieved an R^2 of ____ on the test set, meaning it explains ____% of the variance in house prices, with an average prediction error (MAE) of about \$____. This establishes a working baseline; the improvement ideas above outline clear next steps to boost accuracy further.